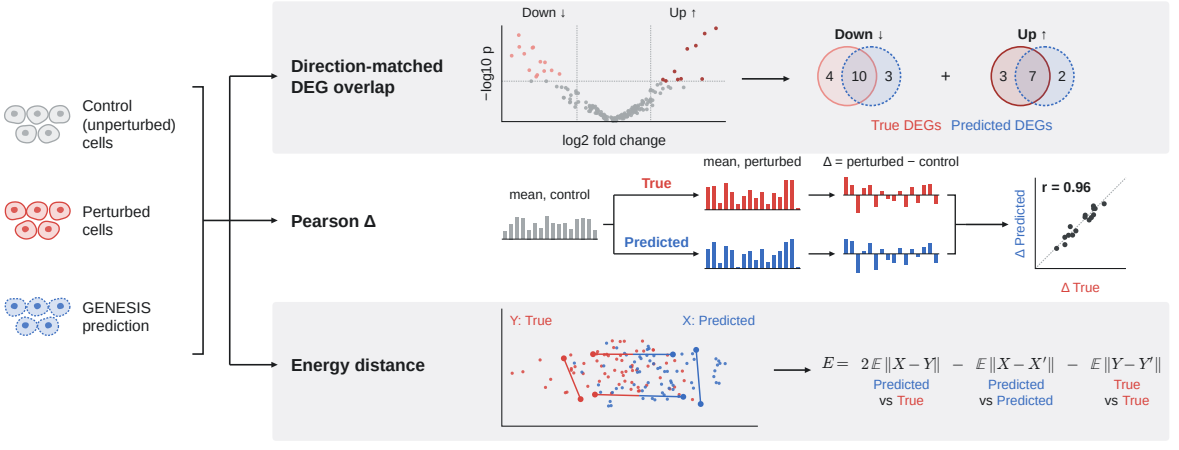

Control (unperturbed) cells
Perturbed cells
GENESIS prediction
Direction-matched DEG overlap
Down ↓
Up ↑
−log10 p
log2 fold change
Down ↓
Up ↑
4
10
3
+
3
7
2
True DEGs
Predicted DEGs
Pearson Δ
mean, control
True
Predicted
mean, perturbed
Δ = perturbed − control
r = 0.96
Δ Predicted
Δ True
Energy distance
Y: True
X: Predicted
$E = 2\mathbb{E}\|X-Y\| - \mathbb{E}\|X-X'\| - \mathbb{E}\|Y-Y'\|$
Predicted vs True
Predicted vs Predicted
True vs True